THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 04

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 1331	$11 \times 11 \times 11 = 1331$
2	**(d)** 512	$8^3 = 512$
3	**(c)** $\frac{8}{125}$	$\frac{2^3}{5^3}$
4	**(d)** Aryabhata	Aryabhatiya
5	**(c)** 5 cm	Diagonals bisect at right angles; use Pythagoras
6	**(a)** square	Both properties
7	**(d)** Yes	Digit sum 18 is divisible by 9
8	**(d)** 81	$a^2 + 2ab + b^2$
9	**(a)** ₹ 150	₹ 15 per pen
10	**(c)** 5 : 2	100 cm : 40 cm
11	**(c)** 1640	2000 × 82/100
12	**(c)** 60	7.5/100 × 800
13	**(c)** 24 cm	$\sqrt{25^2 - 7^2} = 24$
14	**(c)** three numbers satisfying $a^2 + b^2 = c^2$	Definition
15	**(c)** 450 km	Multiply by the scale
16	**(d)** 27	Each stage multiplies the count by 3: $3^3 = 27$
17	**(b)** median	Median
18	**(c)** 19	Top = a + 2b + c
19	**(c)** equal area	Same base and height
20	**(d)** 6 cm²	Right-angled: ½×3×4

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$10^{5-2-2} = 10^1 = 10$ (2).	2
22	$40 + 4 = 44$ (2).	2
23	Yes; 3 and 4 are co-prime and $3 \times 4 = 12$ (2).	2
24	6 cups (2).	2
25	$25 + 36 = 61 \neq 64$, no (2).	2
26	4 hours (2).	2
27	Values far from the rest (1); 54 in 16, 18, 20, 22, 54 (1).	1+1
28	35000 cm² (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Side $= \sqrt{1296} = 36$ m (2); perimeter $= 4 \times 36 = 144$ m (1).	2+1
30	A rhombus (1); each side is half a diagonal and diagonals are equal (2).	1+2
31	$x^2 + 7x + 12$ (2); at $x = 1$: $(4)(5) = 20 = 1 + 7 + 12$ (1).	2+1
32	72% (2); $\frac{18}{25}$ (1).	2+1
33	Cube net has six equal squares; cuboid net has rectangles of three pairs (3).	3
34	$20 \times 21 = 420$ (2); nearest equal pair (1).	2+1
35	$225 + 400 = 625$ (2); area 150 cm² (1).	2+1
	OR: Triplet $(5, 12, 13)$ (1.5); $25 + 144 = 169 = 169$ (1.5).	1½+1½

36	Area $= \frac{1}{2} \times 22 \times 15 = 165$ m² (2); cost ₹ 1980 (1).	2+1
	OR: Other diagonal 14 cm (1.5); side $\sqrt{12^2 + 7^2} \approx 13.9$ cm (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) 4.8×10^7 (1). (ii) $4.8 \times 10^7 + 5 \times 10^6 = 5.3 \times 10^7$ (2). (iii) 5.3×10^7 (2).	1+2+2
	OR: (i) $10^{-5}/10^{-10} = 10^5$ times (3). (ii) 7×10^{-3} mm (2).	3+2
38	(i) ₹ 18000 (2). (ii) ₹ 68000 (1). (iii) ₹ 1888.89 (2).	2+1+2
	OR: (i) 24 girls, 6 with bicycles (3). (ii) 15% (2).	3+2
39	(i) Both 74 (2). (ii) A (1). (iii) 8 and 58 (2).	2+1+2
	OR: (i) $4 \times 12 = 48$; $48 - 32 = 16$ (3). (ii) Sorted 8, 10, 14, 16; median $(10 + 14)/2 = 12$ (2).	3+2
40	(i) $40 + 18 = 58$ units² (3). (ii) Subtract a missing rectangle from a larger one (2).	3+2
	OR: (i) Four congruent right triangles (3). (ii) 16 cm (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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